

Gorman: $V(p, w)$

$$= f(p) + g(p) \cdot w$$

$$u(x) = \min_p V(p, 1) \text{ s.t. } p \cdot x \geq 1$$

WARP: $x, y \in B, x \in C(B), y \not\in C(B)$, then $y \in C(B)$.

WARP $\uparrow$ C₁ Singleton $\uparrow$ WA $\leftarrow$ CLD $\leftarrow$ {WL, HDO} $\leftarrow$ SLP, w NSD $\leftarrow$ SLP, w NSD $\leftarrow$ SLP, w NSD

Rationalizability: $\uparrow$ R₁ Uniqueness $\downarrow$ Unique

$\exists$ complete & transitive $\approx$ Lexicographic Continuity $\rightarrow$ SLP, w NSD

Represented by $\rightarrow$ SLP, w NSD

Convex $\approx$ Quasiconcave $u \rightarrow$ Strict convex $\approx$ Strict quasiconcave $u \rightarrow$ Monotonic $\approx$ Increasing $u \rightarrow$ WL $x(p, w)$

SA: $x, y \in B, x \in C(B), y \not\in C(B)$, then $y \in C(B)$

SARP: $x \in C(B), y \not\in C(B), y \not\geq^I x$, then $p \cdot y > w$

GARP: $x \in C(B), y \not\in C(B), y \not\geq^I x, y \neq x$ then $p \cdot y > w$

$\times$ Integrability: $\rightarrow$ Entire demand

$\rightarrow$ WL & HDO

$\rightarrow$ SARP or NSD & Sym.

$\rightarrow$ Lipschitz continuity $\rightarrow$ Uniqueness

$\rightarrow$ Finite data $\{x_i(p_i, w_i)\}_{i=1}^n$

$\rightarrow$ WL

$\rightarrow$ SARP, GARP

$\rightarrow$ Complete & transitive

$\rightarrow$ Continuous

$\rightarrow$ Monotone

$\rightarrow$ Concave

$\rightarrow$ Unique

Envelope

$$h(p, w) = D_p e$$

$$= D_p e$$

$$= h(p, V(p, w))$$

$$= D_p e$$

$$= D_p e$$

$$= D_p e$$

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Kuhn-Tucker: $x[\nabla u(x) - \gamma p] = 0$

$$x \geq 0, \nabla u(x) - \gamma p \leq 0$$

$$S(p, w) = D_p h(p, w), w = e(p, w)$$

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Choice structure $(B, c) : \forall B \in \mathcal{B} . \emptyset \neq c(B) \subseteq B$. $x \succ y \Leftrightarrow x \succ y$. $x = (0,1)$ $c(x) = \emptyset$
 WA: $\xrightarrow{\text{Richness}} \text{Unique} \xrightarrow{\text{Rationalizable}} \exists ! \text{ complet \& transitive } \succeq \text{ s.t. } \forall B \in \mathcal{B} . c(B) = C_{\succeq}(B)$.
 trivial if $c(B) \subseteq c_{\succeq}(B)$

WA: $\forall x, y \in B_x \cap B_y . x \in c(B_x) . y \in c(B_y) \Rightarrow y \in c(B_x) . x \in c(B_y)$. $B \cap B' = \{y\}$

$\Leftrightarrow \forall x, y \in B_x \cap B_y . x \in c(B_x) . y \notin c(B_x) \Rightarrow y \notin c(B_y)$.

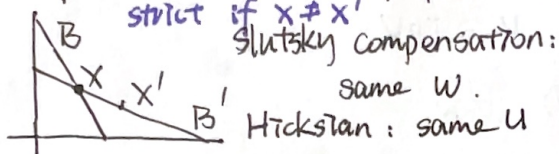
$\Leftrightarrow \forall x, p, w, x', p', w' . x \in c(p, w) . x' \in B . x \neq x' \Rightarrow x' \notin B'$

Only for singleton $c_{\succeq}(p, w)$

3-Dim counterexample.

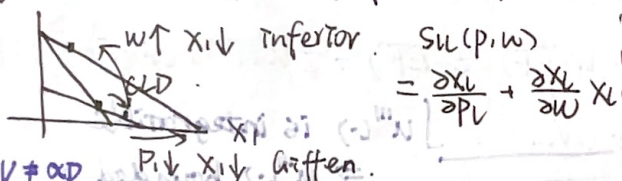
WA
 $\Downarrow$
 CLD

CLD: $x \cdot p' = w' \Rightarrow (x - x') \cdot (p - p') \leq 0$



SE of good 1 wrt $p_1 \leq 0 \Leftrightarrow S_{11}(p, w) \leq 0$.

A good can be Giffen only if it is inferior



CLD

$\Downarrow$
 $\forall v \neq x p$
 $v^T S v < 0$

Slutsky matrix: $S(p, w) = D_p x(p, w) + D_w x(p, w) x^T(p, w)$

NSD: $S_{ik}(p, w) = \frac{\partial x_i}{\partial p_k}(p, w) + \frac{\partial x_i(p, w)}{\partial w} x_k(p, w)$

Symmetric when $L=2$ $S(p, w) p = 0$ & $p^T S(p, w) = 0$.

SA: $\forall x, y \in B . x \in c(B) . y \succeq^{**} x \Rightarrow y \in c(B)$.

UMP: $V(p, w) = \max_x u(x) \text{ s.t. } x \in B(p, w)$.

$D_p V(p, w) = -D_w V(p, w) \cdot x(p, w)$.

$\exists \pi \geq 0 . D_x u(x) \leq \pi p . x > 0 . "$

EMP: $e(p, u) = \min_x p \cdot x \text{ s.t. } u(x) \geq u$.

$D_p e(p, u) = h(p, u)$

$D_p h(p, u) p = 0$

Slutsky: $D_p h(p, u) = D_p x(p, e(p, u)) = D_p x(p, w) + D_w x(p, w) x^T(p, w) = S(p, w)$.

NSD & Symmetric!

$u(x) = \min_p V(p, 1) \text{ s.t. } p \cdot x \geq 1$

$u(x) = \max_u u \text{ s.t. } p \cdot x \geq e(p, u)$

$V_u = \{x : p \cdot x \geq e(p, u) \forall p\}$.

$u(x) = \sup_u \{u : x \in V_u\}$.

Substitutes: $\frac{\partial h_i}{\partial p_k} \geq 0$.

Complements: $\frac{\partial h_i}{\partial p_k} \leq 0$.

Homothetic $\hat{x} \Leftrightarrow HD \perp u(\cdot) \Rightarrow$ linear Engel curve

Quasilinear $\hat{x} \Leftrightarrow u(\cdot) = x_1 + \phi(x_2, \dots, x_n)$

$w \uparrow x_1 \uparrow$

WE: $\frac{\partial x}{\partial w}(p, w)$ Normal ≥ 0 Inferior < 0 .

PE: $\frac{\partial x}{\partial p}(p, w)$ Giffen $\frac{\partial x_i}{\partial p_i} > 0$

WL: $x^T(p, w) p = w$.

$\Rightarrow \sum_k \frac{\partial x_k}{\partial p_k} p_k + x_k = 0$ At least 1 non-Giffen good.

$\Rightarrow \sum_k \frac{\partial x_k}{\partial w} p_k = 1$ At least 1 normal good

HDO: $x(p, w) = x(\pi p, \pi w)$

$\Rightarrow \sum_k \frac{\partial x_k}{\partial p_k} p_k + \frac{\partial x_k}{\partial w} w = 0$

SARP: $x \succeq^{in} y . y \neq x$.

$\downarrow y \in c(B(p, w)) \Rightarrow p \cdot x > w$.

GARP: $x \succeq^{in} y . y \neq x$.

$y \in c(B(p, w)) \Rightarrow p \cdot x \geq w$

$\succeq^{**}$ revealed indirectly preferred.

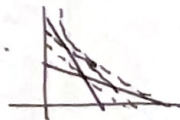
$x(p, w)$: WL. HDO. Convex / Unique

$V(p, w)$: $\uparrow w . \downarrow p$. HDO. quasiconvex continuous.

$p h(p, u) = e(p, u)$

$h(p, w)$: Tight: HDO, CLD.

$e(p, u)$: $\uparrow p - \uparrow u$. HD $\perp$. concave.

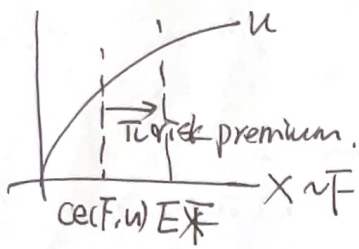


VNM Axiom: ① Order (complete & transitive) ② Mixture continuity ③ Independence.

finite $\mathbb{Z}$.

② + ③ $\Leftrightarrow \forall L' > L > L'' \exists \alpha \in (0,1)$
 $s.t. L \sim \alpha L' + (1-\alpha) L''$

VNM Expected Utility: $\exists u: \mathbb{Z} \rightarrow \mathbb{R}$ s.t. $\forall L^F, U(L^F) = \int u(x) dF(x)$.



Risk Averse: $ce(F, u) \leq EF, \forall F$.

$\Leftrightarrow u(\cdot)$ concave.

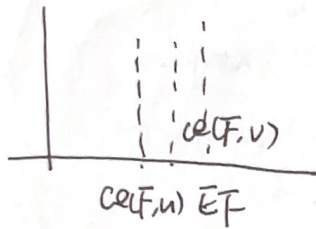
$\Leftrightarrow \pi(F) \geq 0, \forall F$.

$\succeq^u$ is more RA than $\succeq^v$. ① $S_{const} \succeq^v F \Rightarrow S_{const} \succeq^u F$.

② $ce(F, v) \geq ce(F, u)$. ③ $\pi(F, v) \leq \pi(F, u)$.

④ $\exists f(\cdot)$ concave s.t. $u = f \circ v$.

⑤ $r_A(x, u) \geq r_A(x, v) \forall x$.



Consider a lottery F with domain $[x-\varepsilon, x+\varepsilon]$. ε is small enough. $\forall x$.

$$U(ce(F, u)) = U(EF - \pi(x, u)) = \int_{x-\varepsilon}^{x+\varepsilon} u(z) dF(z)$$

$$\Rightarrow U(EF) - U'(EF) \pi(x, u) + o(\pi) = \int u(EF) + u'(EF)(z-EF) + \frac{1}{2} u''(EF)(z-EF)^2 + o(z^2) dF(z)$$

$$\Rightarrow -U'(EF) \pi(x, u) + o(\pi) = \frac{1}{2} u''(EF) \text{Var} F + \frac{d(\text{Var} F)}{2} \int u'''(\cdot) \text{ is integrable}$$

$$\Rightarrow \boxed{\pi(x, u) + o(\pi)} = -\frac{1}{2} \text{Var} F \frac{u''(EF)}{u'(EF)} =: \boxed{\frac{\text{Var} F}{2} r_A(EF, u)}$$

$\Rightarrow u'''(\cdot)$ bounded
 e.g. $\ln(\cdot)$ unbounded at $\cdot=0$

F on $[x(1-\varepsilon), x(1+\varepsilon)]$: $\boxed{\frac{\pi(x, u)}{x}} = -\frac{1}{2} \text{Var} F \frac{u''(EF)}{u'(EF)} =: \boxed{\frac{\text{Var} F}{2} r_R(EF, u)}$

therefore, $EF = EG$. $\text{Var} F > \text{Var} G$

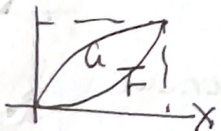
$\Rightarrow F \prec G$

CRRA: $u(x) = \frac{C_1 x^{1-R} + C_0}{1-R}, R \neq 1, u(x) = \ln x, R=1$.

CARA: $u(x) = C_0 e^{-R_A x} + C_1$



FOSD: $F \succeq^{FOSD} G$.



① $\forall$ increasing $u: \int u(x) dF(x) \geq \int u(x) dG(x)$.

② $\forall x, F(x) \leq G(x)$.

③ F is a "shifting right transform" of G .

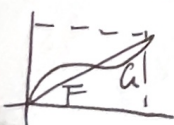
SOSD: $F \succeq^{SOSD} G$.

① $\forall$ increasing, concave $u: \int u(x) dF(x) \geq \int u(x) dG(x)$.

② $\forall x, \int^x F(t) dt \leq \int^x G(t) dt$.

③ G is a "Mean-preserving spread" of F .

stochastic
 $F \xrightarrow{H} G$



Savage's Subjective Expected Utility.

$$U(f) = \int_S u \circ f d\mu = \sum_S \mu(S) u(f(S)) : \mu \in \Delta S, \mu: S \rightarrow \mathbb{R} \text{ measure "Subjective Prob"}$$

$$f: S \rightarrow \mathbb{Z} \quad u: \mathbb{Z} \rightarrow \mathbb{R} \text{ index utility}$$

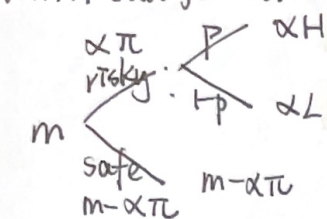
Maximin Model:

$$U(f) = \min_{\mu \in P} \int_S u \circ f d\mu. \quad P \subseteq \Delta S \text{ set of subjective probability.}$$

Motivation: under VNM EU: increasing $U(\cdot) \Rightarrow$ demand for risky asset > 0 . ^{without} knife edge

Evidence from empirical research: $\exists$ Non-participation!

VNM / Savage EU:



$$U(\alpha, p) = p U(\alpha H - \alpha \pi + m) + (1-p) U(\alpha L - \alpha \pi + m).$$

p : objective or subjective.

$$H > \pi > L.$$

$$= p U(m_H) + (1-p) U(m_L).$$

$$\begin{cases} m_H(\alpha) = \alpha(H - \pi) + m \\ m_L(\alpha) = \alpha(L - \pi) + m \end{cases} \Rightarrow \frac{m_H(\alpha) - m}{m_L(\alpha) - m} = \frac{H - \pi}{L - \pi}$$

|slope| of BL = $\frac{H - \pi}{L - \pi}$

$$\text{FOC from } U(\alpha; p): \frac{\partial U}{\partial \alpha}(\alpha, p) = p(H - \pi) u'(m_H) + (1-p)(L - \pi) u'(m_L) = 0$$

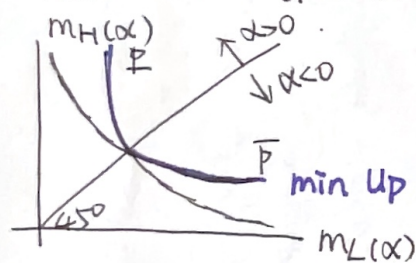
$$\Rightarrow \frac{H - \pi}{\pi - L} = \frac{(1-p) u'(m_L)}{p u'(m_H)} \quad \text{BL (slope) = MRS.}$$

If $\alpha = 0$ Nonparticipation is optimal: $m_H = m_L = m \xrightarrow{\text{All DM}} \pi = pH + (1-p)L$ fairly priced.

If $\alpha > 0$ is optimal: $m_H > m_L, u'(m_L) > u'(m_H) \Rightarrow \pi < pH + (1-p)L$ under priced. ^{RA DM}

If $\alpha < 0$ is optimal: $m_L > m_H, u'(m_H) > u'(m_L) \Rightarrow \pi > pH + (1-p)L$ over priced.

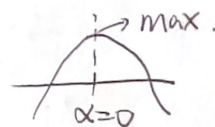
$$\text{Maximin EU: } U(m_L, m_H) = \min_{p \in [\underline{p}, \bar{p}]} \{ p U(m_H) + (1-p) U(m_L) \}$$



$$= \min_{p \in [\underline{p}, \bar{p}]} \{ p [U(m_H) - U(m_L)] + U(m_L) \}$$

$$= \begin{cases} \underline{p} [U(m_H) - U(m_L)] + U(m_L) & \text{if } m_H > m_L \\ \bar{p} [U(m_H) - U(m_L)] + U(m_L) & \text{if } m_H \leq m_L \end{cases}$$

$$\text{FOC: } \frac{\partial U}{\partial \alpha} = \begin{cases} \underline{p} (H - \pi) u'(m_H) + (1 - \underline{p}) (L - \pi) u'(m_L) & \text{if } \alpha > 0 \\ \bar{p} (H - \pi) u'(m_H) + (1 - \bar{p}) (L - \pi) u'(m_L) & \text{if } \alpha < 0 \end{cases}$$



If $\alpha = 0$ Nonparticipation is optimal: $m_H = m_L = m$ & $\frac{\partial U}{\partial \alpha}(\underline{p}) < 0$ & $\frac{\partial U}{\partial \alpha}(\bar{p}) > 0$
 $\Rightarrow \pi \in [\underline{p}H + (1 - \underline{p})L, \bar{p}H + (1 - \bar{p})L]$

If $\alpha > 0$ is optimal: $\pi < \underline{p}H + (1 - \underline{p})L$ pessimistic forecast.

If $\alpha < 0$ is optimal: $\pi > \bar{p}H + (1 - \bar{p})L$ positive forecast.